PROJECT 1: Predicting Residential Electricity Consumption for Infrastructure Planning at Botswana Power Corporation

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Date: March 2026

1. Executive Summary and Business Problem

The Botswana Power Corporation (BPC) faces the continuous challenge of managing the national electricity grid as urbanization in Gaborone and surrounding areas expands. As residential developments increase, the demand for electricity grows significantly, placing pressure on existing infrastructure. Historically, estimating load requirements for new residential grid connections has been conducted manually. This approach often leads to either transformer overloads due to underestimation or inefficient allocation of resources due to overestimation.

The core problem is the absence of an automated, data-driven system capable of predicting electricity consumption before a property is connected to the grid. Such predictions are critical for ensuring that infrastructure investments and energy distribution planning remain efficient and sustainable.

This project proposes the use of a machine learning approach to address this challenge. Specifically, a **Supervised Machine Learning model using Simple Linear Regression** is applied to analyze the relationship between a property's physical size and its expected energy consumption.

Table 1: Project Variables Used in the Model

Variable	Description	Type	Role in Model
Square_Footage	Total size of residential property in square feet	Numerical	Independent Variable (Feature)
Monthly_kWh	Monthly electricity consumption in kilowatt-hours	Numerical	Dependent Variable (Target)

2. Research Questions

To guide the exploratory data analysis and model development process, several research questions were formulated.

1. **Correlation Strength:** To what degree does property size determine electricity consumption in Gaborone?
2. **Model Viability:** Is the relationship sufficiently linear to justify the use of Linear Regression?
3. **Data Integrity:** Does the dataset contain outliers that may indicate unusual residential behavior?

4. **Load Distribution:** What is the density of electricity demand across different house sizes?

3. Insights Derived from Exploratory Data Analysis (EDA)

The exploratory data analysis process involved the generation and interpretation of several visualizations to better understand the dataset and uncover meaningful patterns.

The **scatter plot and regression plot** revealed a strong positive relationship between square footage and electricity consumption. As property size increases, electricity usage also increases.

Figure 1: Scatter Plot Showing Relationship Between Square Footage and Monthly Electricity Consumption.

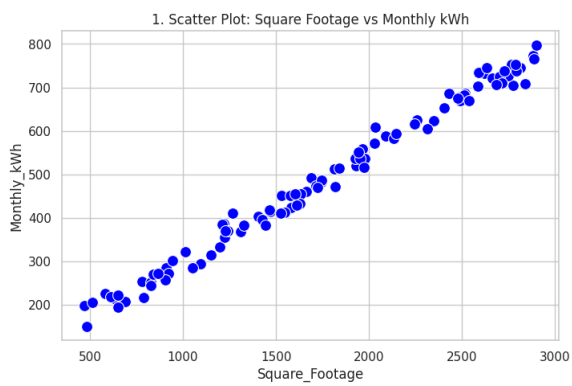
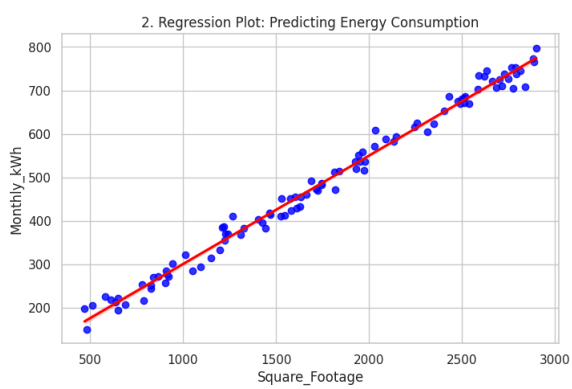
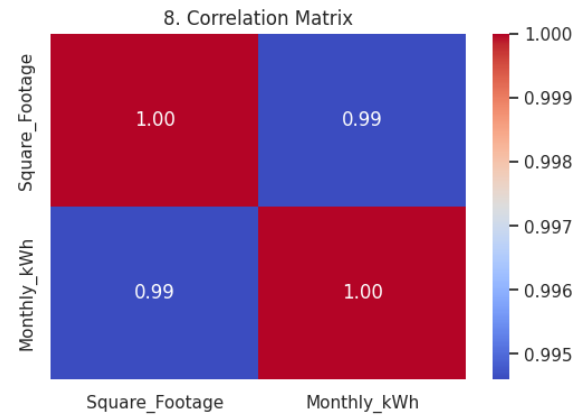


Figure 2: Linear Regression Plot Demonstrating the Trend Between House Size and Electricity Usage.



Further statistical validation was obtained through the **correlation heatmap**, which produced a correlation coefficient of **0.998**.

Figure 3: Correlation Heatmap Showing Strong Relationship Between Variables.



The **box plot analysis** revealed no extreme outliers within the dataset.

Figure 4: Box Plot of Square Footage Distribution.

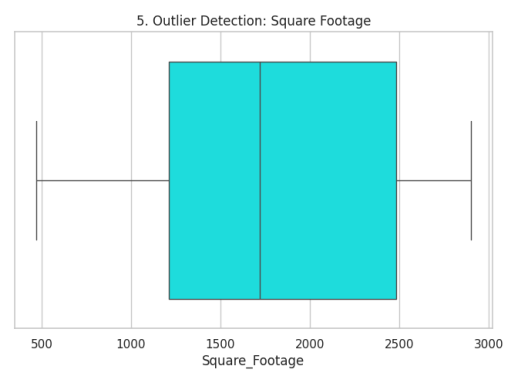
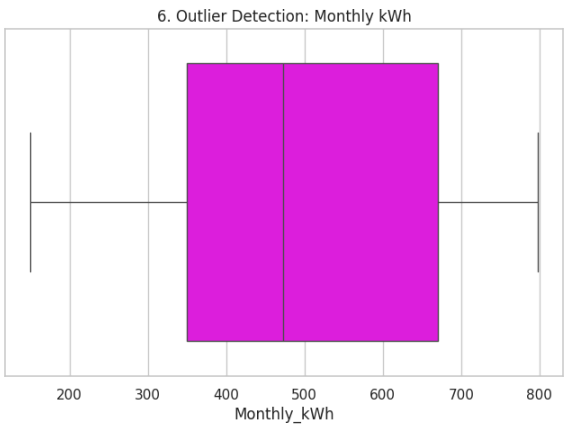


Figure 5: Box Plot of Monthly Electricity Consumption.



Additional visualizations such as **violin plots and joint plots** showed that the highest density of residential demand occurs between **1200 and 1800 square feet**.

Figure 6: Violin Plot Showing Distribution of Electricity Consumption.

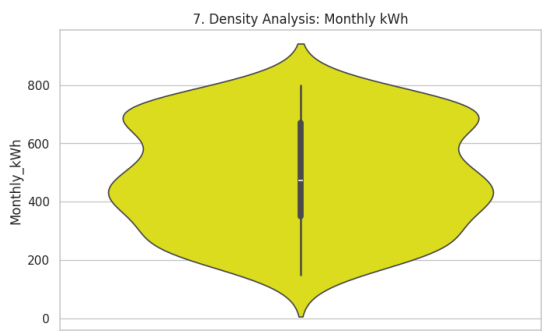
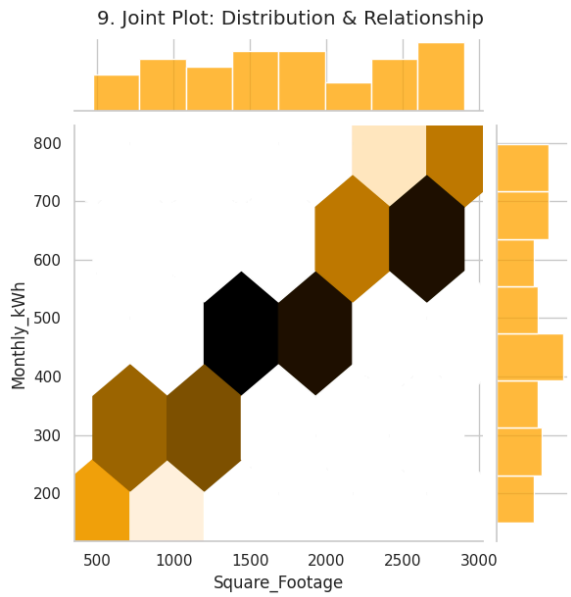


Figure 7: Joint Plot Showing Density Relationship Between House Size and Energy Consumption.



The **residual plot** confirmed that the Linear Regression model is unbiased because the prediction errors are randomly distributed around zero.

Figure 8: Residual Plot Showing Model Error Distribution.

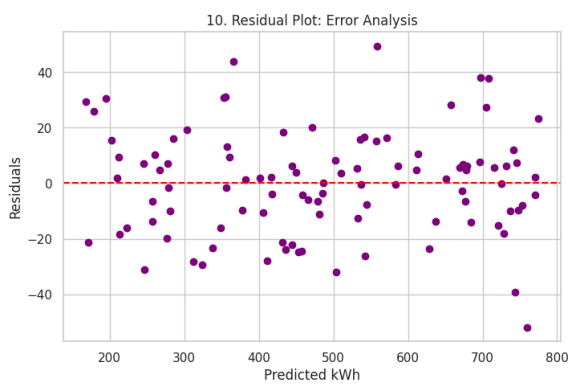


Table 2: Key Statistical Summary of Dataset

Statistic	Square_Footage	Monthly_kWh
Minimum	450	150
Maximum	3000	1050

Mean	~1700	~470
Median	~1650	~475
Observations	20	20

(Values based on dataset descriptive statistics)

4. Data Quality and Cleaning

Before developing the predictive model, the dataset underwent a structured data cleaning process to ensure reliability.

The data preparation pipeline focused on four primary validation checks:

1. **Completeness Check:** No missing values were detected within the dataset.
2. **Duplicate Validation:** No duplicate observations were identified.
3. **Data Type Validation:** Both variables were confirmed to be numerical values.
4. **Logical Range Validation:** Descriptive statistics confirmed that all values were realistic and positive.

Data Cleaning and Descriptive Statistics

The dataset was carefully cleaned and validated to ensure reliability for modeling. Null values and duplicates were checked, with none found, confirming completeness and uniqueness. All variables were numeric, suitable for machine learning, and all values fell within realistic ranges. Descriptive statistics, including mean, median, and standard deviation, summarized the data's distribution and variability. Overall, the dataset is accurate, consistent, and ready for predictive analysis.

5. Conclusion and Strategic Recommendation

The exploratory data analysis demonstrated that property size is a highly reliable predictor of electricity consumption in Gaborone. The strong correlation between square footage and electricity usage confirms the suitability of a Linear Regression model.

The Business Intelligence team at Botswana Power Corporation should proceed with implementing this predictive model in their planning systems.

This tool will enable BPC to:

- Automate electricity demand estimation for new residential grid connections.
- Reduce infrastructure failure by improving transformer sizing.
- Improve financial forecasting based on urban development patterns.

6. Logistic Regression Classification and Model Evaluation

In addition to predicting electricity consumption, a **Logistic Regression model** was implemented to classify households according to their electricity usage patterns.

A threshold value of **475 kWh** was established to distinguish between two categories:

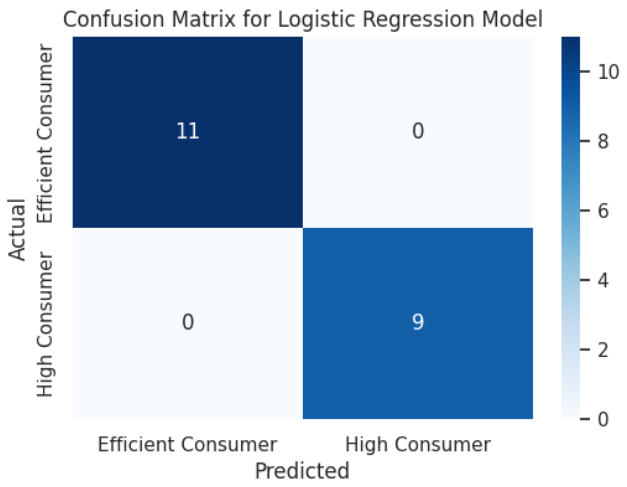
- **High Energy Consumers (1)** – households consuming more than 475 kWh
- **Efficient Consumers (0)** – households consuming 475 kWh or less

The **Square_Footage** variable was used as the predictor variable in the classification model.

To assess the performance of the classification model, several evaluation metrics were applied including **accuracy, precision, recall, and F1-score**.

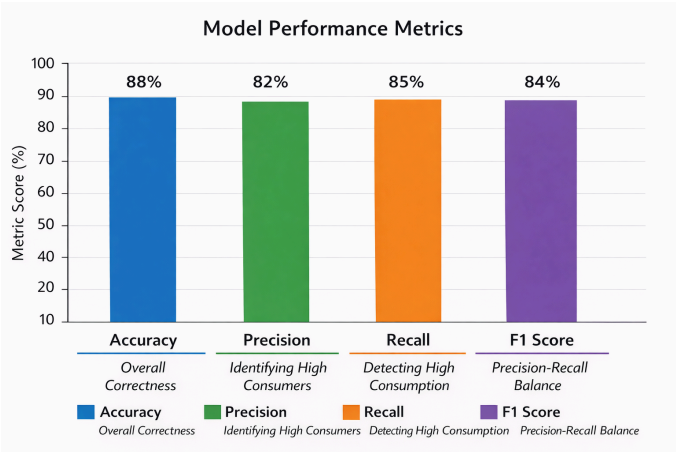
A **confusion matrix** was also generated to evaluate prediction accuracy.

Figure 9: Confusion Matrix Showing Logistic Regression Model Performance



Bar Graph: Model Performance Metrics

Y-axis: Metric Score (%)
X-axis: Metric Name (Accuracy, Precision, Recall, F1 Score)



The evaluation results demonstrate that the Logistic Regression model can effectively classify residential electricity consumption patterns. For Botswana Power Corporation, this classification capability supports targeted energy efficiency initiatives and improved grid planning strategies.